

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Aim : Scenario-Based Case Study Questions

Theory :

Case Study 1 — The Fast Startup

A startup builds a food delivery app using LLM tools. The team generates most backend APIs using AI. The app launches quickly but after 2 months becomes hard to update. Developers struggle to understand dependencies.

1. Problem:

- Lack of maintainability and accumulation of technical debt

2. Violated SE Principle:

- Modularity & Maintainability principle
- Also violated proper documentation practices

3. Preventive Practices:

- Maintain proper documentation of APIs and dependencies
- Conduct regular code reviews and refactoring

Case Study 2 — Banking Failure

A bank integrates an AI-generated fraud detection module. The code passes all tests but blocks legitimate transactions during a festival sale.

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1. Testing Limitation:

- Failure to handle edge cases / real-world scenarios

2. Additional Testing Approach:

- User Acceptance Testing (UAT)
- Domain-specific testing (real transaction scenarios)
- Stress & scenario-based testing

3. Responsibility:

- Both developer and company are responsible
- Justification:
 - Developers must validate AI output
 - Company must ensure domain correctness and safety
 - AI provider is not fully responsible as it is only a tool

Case Study 3 — The Lazy Team

Students build a mini-project entirely using ChatGPT prompts. During viva they cannot explain class relationships.

1. Failed Learning Outcome:

- Lack of conceptual understanding

2. Why AI is Dangerous Without Validation:

- Students may rely on output without learning logic
- Leads to surface-level knowledge only

3. New Evaluation Method:

- Conduct oral viva + live coding
- Use concept explanation-based evaluation

Case Study 4 — Agile Team with AI Manager

A company uses AI to assign tasks and estimate sprint effort. Team productivity increases but developers feel disengaged.

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1. Affected Agile Principle:

- Individuals and interactions over processes and tools

2. Balance Between AI & Humans:

- AI should assist, not replace decisions
- Humans must handle:
 - creativity
 - collaboration
 - final decision-making

3. Future of Agile:

- Agile will evolve, not disappear
- Will include AI-assisted planning and estimation

Case Study 5 — Safety Critical System

An autonomous vehicle company uses AI-generated control software. The software works 99.9% correctly but fails rarely in unpredictable situations.

1. Why Unacceptable:

- Even 0.1% failure can cause loss of life
- Safety systems require near 100% reliability

2. Additional Verification:

- Formal verification
- Rigorous testing & simulation
- Certification standards (e.g., safety compliance)

3. Should AI Code Be Allowed?

- Yes, but only with:
 - strict human validation
 - formal testing
 - regulatory approval

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Exam Questions

1. The Identity Question

If an AI system designs architecture, generates code, writes tests and fixes bugs — at what point does the human stop being the developer and start being only a supervisor? Is that still Software Engineering or a new discipline?

- Human becomes a supervisor / system designer
- Still Software Engineering, but evolving into:
 - AI-assisted Software Engineering
- Focus shifts from coding → design, validation, decision-making

2. Responsibility & Accountability

When an AI-generated module causes a critical failure in a banking or medical system, who should be legally and ethically responsible — the developer, the company, or the AI provider?

- Primary responsibility: Company
- Secondary: Developers
- AI provider:
 - only responsible for tool reliability
- Ethical conclusion:
 - Humans must remain accountable

3. Skill Shift Problem

In a world where LLMs can generate syntactically correct code instantly, what becomes more valuable: knowing programming languages or understanding problem abstraction? Justify with future industry roles.

- More valuable skill:
 - **Problem abstraction & system design**

Future roles:

- AI-assisted developer

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- System architect
- AI validator / reviewer

4. The Debugging Paradox

AI can write code faster than humans can understand it. Will future engineers spend more time creating software or understanding software? Explain the long-term impact on maintenance engineering.

- Engineers will spend more time:
→ Understanding & debugging software

Impact:

- Increase in maintenance engineering importance
- Code comprehension becomes critical skill

5. Creativity vs Automation

If millions of developers use similar LLM models trained on the same datasets, will software systems become standardized and less innovative? Could AI unintentionally reduce creativity in software design?

Risk:

→ Software becomes standardized

However:

→ Creativity will shift to:

- architecture
- problem-solving
- innovation layers

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6. Agile Methodology Relevance

Agile assumes humans estimate effort and uncertainty. If AI can predict sprint timelines and generate features automatically, does Agile methodology still make sense, or will a new development process replace it?

Agile will not disappear

It will transform into:

→ AI-augmented Agile

Humans still needed for:

- requirement changes
- stakeholder communication

7. Learning Crisis in Education

If students rely heavily on AI to generate assignments and projects, can they still become engineers? How should Software Engineering education change in the next 5 years?

- Students may not become true engineers if overdependent on AI

Required changes:

- More practical + viva-based evaluation
- Focus on:
 - logic
 - understanding
 - real problem solving

8. Security & Trust Question

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AI models generate code they do not truly “understand.” Should AI-generated code be trusted in safety-critical systems like aircraft, autonomous vehicles, or surgery robots? Why or why not?

- AI-generated code should NOT be blindly trusted

Reason:

- AI lacks true understanding
- May introduce hidden bugs or vulnerabilities

Conclusion:

→ Use AI with strict verification in critical systems

9. The End of Programming Languages?

If future developers interact with computers using natural language instead of programming languages, will programming languages disappear — or evolve into something else?

- Programming languages will not disappear
- They will evolve

Future:

- Natural language → high-level instructions
- Programming languages → execution layer

10. Future Role Question (Most Important)

Predict the job description of a Software Engineer in 2035. List 5 skills that will disappear and 5 skills that will become essential because of LLMs like ChatGPT and Anthropic.

Job Description (2035):

- Design system architecture
- Use AI tools to generate code
- Validate and test AI outputs

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- Ensure security, ethics, and reliability
- Collaborate with AI systems

5 Skills That May Disappear

1. Manual syntax memorization
2. Writing boilerplate code
3. Basic debugging
4. Repetitive coding tasks
5. Low-level implementation focus

5 Skills That Will Become Essential

1. Problem abstraction & system design
2. AI tool usage & prompt engineering
3. Code validation & debugging skills
4. Critical thinking & decision-making
5. Security and ethical awareness

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